

SOAP2REST Bridge

Technical Requirements Document (TRD)

Project Overview

SOAP2REST Bridge is an AI-powered platform that converts legacy SOAP/XML services into modern REST/JSON APIs. The system parses WSDL files, generates API mappings using AI, and acts as a runtime proxy for JSON-to-XML and XML-to-JSON transformation.

Technology Stack

Frontend: React, Vite, TypeScript, Tailwind CSS, Recharts

Backend: Spring Boot 3.x, Java 21, Maven

Database: H2 (Development), SQLite (Optional)

Authentication: JWT Authentication + API Key Management

Documentation: Swagger/OpenAPI

AI Services: Gemini API / Ollama

Logging: SLF4J + Logback

Architecture Components

1. WSDL Ingestion Engine
2. AI Mapping Agent
3. Dynamic REST Route Generator
4. Runtime Proxy Engine
5. Security Layer
6. Analytics & Monitoring Module
7. Mock SOAP Service

Third-Party APIs

- Gemini API – Semantic Mapping Generation
- Ollama – Local LLM Support
- Swagger/OpenAPI – Documentation Generation

Environment Variables

GEMINI_API_KEY
JWT_SECRET
DB_URL
DB_USERNAME
DB_PASSWORD
SOAP_TIMEOUT
API_KEY_EXPIRY_DAYS

Folder Structure

soap2rest-bridge/
■■■ backend/
■■■ frontend/
■■■ docs/
■■■ scripts/
■■■ deployment/

Constraints

- Must support SOAP 1.1 and SOAP 1.2
- Must generate REST APIs dynamically
- Must support JWT and API Key security
- Must be responsive on desktop and mobile
- Must provide analytics and monitoring
- No hardcoded secrets in source code